

TIDES: Three-Layer Inference Architecture

Algorithm Instruction

Working specification for the redesigned TIDES inference pipeline

1 Overview

TIDES organizes temporal network inference into three consecutive layers:

Temporal segmentation $\longrightarrow$ Edge-space reconstruction $\longrightarrow$ Physical compression

The three layers answer three distinct questions:

- Step 1:** When does the vector field change?

Step 2: Which edge-space vector fields are compatible with the trajectory?

Step 3: Which physical topology–dynamics representation gives the shortest description?

The complete inference chain is

$$x(t) \longrightarrow \tau_{1:K-1} \longrightarrow \mathcal{B}_\varepsilon \longrightarrow \mathcal{H}^\varepsilon \longrightarrow (W, \Theta)_H^*$$

Step 1 extracts temporal structure from the observed trajectory. Step 2 reconstructs the full family of edge-space vector fields compatible with the observations. Step 3 introduces a physical hypothesis H , factorizes the edge-space family into topology and dynamics, and uses minimum description length (MDL) to select the shortest admissible representation.

The central representation hierarchy is

$$x(t) \longleftarrow B_{1:K} \longleftarrow (W, \Theta),$$

and TIDES performs inference in the reverse direction:

trajectory $\longrightarrow$ edge-space solution family $\longrightarrow$ physical representation family
 $\longrightarrow$ minimum-description representation

2 Local Interaction Library and Edge-Space Lifting

Consider a network with N nodes and M candidate edges. The local interaction law is represented in a prescribed library

$$\Psi = \{\psi_1, \dots, \psi_L\}.$$

The corresponding interaction coefficients are collected as

$$\Theta = \{\theta_1, \dots, \theta_L\}.$$

At temporal stage r , both topology and interaction strengths may in general vary. The effective interaction law carried by edge m is written as

$$u_m^{(r)} = w_m^{(r)} \sum_{\ell=1}^L \theta_\ell^{(r)} \psi_\ell.$$

Here $w_m^{(r)}$ is the structural amplitude of edge m , while $\theta_\ell^{(r)}$ is the coefficient of the ℓ -th library function. The library functions ψ_ℓ themselves remain fixed.

Moving the edge amplitude inside the sum gives

$$u_m^{(r)} = \sum_{\ell=1}^L w_m^{(r)} \theta_\ell^{(r)} \psi_\ell.$$

This motivates the lifted coefficients

$$B_{m\ell}^{(r)} = w_m^{(r)} \theta_\ell^{(r)}.$$

Therefore

$$u_m^{(r)} = \sum_{\ell=1}^L B_{m\ell}^{(r)} \psi_\ell.$$

Collecting all lifted coefficients gives the stage-wise edge-space matrix

$$B^{(r)} = \begin{bmatrix} B_{m\ell}^{(r)} \end{bmatrix} \in \mathbb{R}^{M \times L}.$$

The full temporal edge-space field is

$$B_{1:K} = \left(B^{(1)}, \dots, B^{(K)} \right).$$

The lifting

$$(w^{(r)}, \Theta^{(r)}) \mapsto B^{(r)}$$

separates the observational reconstruction problem from the subsequent physical factorization problem. Once lifted, the trajectory is linear in the unknown coefficients $B^{(r)}$.

3 Step 1 - Temporal Segmentation

Given the observed trajectory

$$x(t),$$

Step 1 detects changepoints of the effective vector field,

$$\tau_{1:K-1} = (\tau_1, \dots, \tau_{K-1}).$$

These changepoints partition the trajectory into K temporal stages,

$$I_1, \dots, I_K.$$

The Step-1 map is therefore

$$x(t) \longrightarrow \tau_{1:K-1}.$$

Each stage subsequently provides observational constraints on one stage-wise edge-space field $B^{(r)}$.

4 Step 2 - Edge-Space Vector-Field Family Reconstruction

After lifting, the reconstruction problem is linear in $B^{(r)}$.

For stage r , let

$$b_r = \text{vec}(B^{(r)}).$$

The observational equation is

$$y_r = A_r b_r + \eta_r,$$

where y_r contains the observed vector-field information and A_r is constructed from the observed states and the evaluations of the library functions ψ_ℓ .

Stacking all stages gives

$$y = \begin{bmatrix} y_1 \\ \vdots \\ y_K \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ \vdots \\ b_K \end{bmatrix},$$

and hence

$$y = Ab + \eta.$$

Define the relative reconstruction residual

$$\rho(B_{1:K}) = \frac{\|y - A \text{vec}(B_{1:K})\|_2}{\|y\|_2}.$$

Given the observational uncertainty floor ε , the complete feasible edge-space family is

$$\mathcal{B}_\varepsilon = \{B_{1:K} : \rho(B_{1:K}) \leq \varepsilon\}.$$

Thus Step 2 is

$$(x(t), \tau_{1:K-1}, \Psi) \longrightarrow \mathcal{B}_\varepsilon.$$

4.1 Geometry of the Solution Family

Let the effective singular-value decomposition of the stacked observation operator be

$$A = U_q \Sigma_q V_q^\top,$$

where $q = \text{rank}(A)$. Let

$$b_0 = A^+ y$$

be a minimum-residual reference solution, with residual

$$r_0 = y - Ab_0.$$

Any coefficient vector can be decomposed as

$$b = b_0 + V_q a + V_0 z,$$

where V_0 spans $\ker A$. Since r_0 is orthogonal to $\text{col}(A)$,

$$\|y - Ab\|_2^2 = \|r_0\|_2^2 + \|\Sigma_q a\|_2^2.$$

Hence the feasible family has the geometric form

$$\text{identified-space ellipsoid} \times \text{observational nullspace}.$$

The identifiable directions encode combinations of edge-space coefficients constrained by the trajectory, while the null directions encode observationally unresolved degrees of freedom.

4.2 Temporal-Change View

For any feasible temporal field $B_{1:K}$, define

$$\Delta B^{(r)} = B^{(r+1)} - B^{(r)}, \quad r = 1, \dots, K-1.$$

The temporal-change sequence is

$$\Delta B_{1:K-1} = (\Delta B^{(1)}, \dots, \Delta B^{(K-1)}).$$

Applying this difference operation to the full feasible family gives

$$\Delta \mathcal{B}_\varepsilon = \{\Delta B_{1:K-1} : B_{1:K} \in \mathcal{B}_\varepsilon\}.$$

Thus Step 2 provides two complementary representations of the same observational information:

$$\mathcal{B}_\varepsilon \quad \text{and} \quad \Delta\mathcal{B}_\varepsilon.$$

The first is the absolute edge-space field family; the second is its temporal-change view.

5 Step 3 - Physical Representation under Hypothesis H

A physical hypothesis is denoted by

$$\boxed{H}.$$

The set of all topology–dynamics representations satisfying that hypothesis is denoted by

$$\boxed{\mathcal{H}}.$$

A candidate representation specifies topology coefficients and interaction coefficients and generates a temporal edge-space field

$$B_{1:K}(W, \Theta).$$

The representations compatible with the Step-2 field family are

$$\boxed{\mathcal{H}^\varepsilon = \{(W, \Theta) \in \mathcal{H} : B_{1:K}(W, \Theta) \in \mathcal{B}_\varepsilon\}}.$$

Step 3 therefore searches the physical representation family induced by H inside the observational family reconstructed in Step 2.

5.1 Fixed Dynamics

$$\boxed{H_{\text{FD}} : \quad \Theta^{(1)} = \dots = \Theta^{(K)} = \Theta.}$$

Then

$$\boxed{B^{(r)} = W^{(r)}\Theta^\top},$$

and

$$\boxed{\Delta B^{(r)} = \Delta W^{(r)}\Theta^\top}.$$

The temporal changes share a common dynamical direction Θ , while the structural amplitudes vary with stage.

5.2 Fixed Topology

$$\boxed{H_{\text{FT}} : \quad W^{(1)} = \dots = W^{(K)} = W.}$$

Then

$$B^{(r)} = W \Theta^{(r)\top},$$

and

$$\Delta B^{(r)} = W \Delta \Theta^{(r)\top}.$$

The temporal changes share a common structural direction W , while the interaction coefficients vary with stage.

The same observational objects

$$\mathcal{B}_\varepsilon, \quad \Delta \mathcal{B}_\varepsilon$$

therefore support different hypothesis-dependent factorization paths.

6 Conditional MDL

For a representation

$$(W, \Theta) \in \mathcal{H}^\varepsilon,$$

define the conditional description length

$$L_{\text{MDL}}(W, \Theta \mid H, \Psi) = L_{\text{struct}} + L_{\text{expr}} + L_{\text{prec}}.$$

The selected representation is

$$(W, \Theta)_H^* = \arg \min_{(W, \Theta) \in \mathcal{H}^\varepsilon} L_{\text{MDL}}(W, \Theta \mid H, \Psi).$$

6.1 Structural Code

L_{struct} describes the structural degrees of freedom left unspecified by H .

For H_{FD} , let the changed-edge support at transition k be

$$S_k, \quad E_k = |S_k|.$$

With M candidate edges, an elementary support code is

$$L_{\text{struct}}^{(k)} = \log(M + 1) + \log \binom{M}{E_k}.$$

If the transition supports are encoded independently,

$$L_{\text{struct}} = \sum_{k=1}^{K-1} \left[\log(M + 1) + \log \binom{M}{E_k} \right].$$

Additional structural blocks, such as a baseline support when required by H , are encoded analogously.

6.2 Expression Code

Assume the interaction library is polynomial and that the maximum available polynomial resolution is P . Let

$$\Psi_P = \{\psi_1, \dots, \psi_{L_P}\}.$$

For an independent interaction law

$$f = \sum_{\ell \in J} \theta_\ell \psi_\ell,$$

define

$$s = |J|,$$

and let

$$p = \max_{\ell \in J} \deg \psi_\ell$$

be the largest active polynomial degree. Define

$$L_p = \#\{\psi_\ell : \deg \psi_\ell \leq p\}.$$

A hierarchical code for the active polynomial expression is

$$L_{\text{expr}}(f) = \log P + \log L_p + \log \left[\binom{L_p}{s} - \binom{L_{p-1}}{s} \right].$$

Under H_{FD} , the interaction law is shared across all stages and is therefore encoded once. More generally,

$$L_{\text{expr}} = \sum_a L_{\text{expr}}(f_a),$$

where the sum runs over the independent interaction-law objects implied by H .

6.3 Precision Code

Let Q be the number of free numerical coefficients in a candidate physical representation. For a q -bit quantized representation, define

$$q^* = \min \left\{ q : \exists (\widetilde{W}, \widetilde{\Theta})_q \text{ such that } B_{1:K}(\widetilde{W}, \widetilde{\Theta}) \in \mathcal{B}_\varepsilon \right\}.$$

The numerical precision cost is

$$L_{\text{prec}} = L_{\mathbb{N}}(q^*) + Qq^* \log 2.$$

The total MDL therefore balances

$$\text{structural complexity} + \text{functional complexity} + \text{numerical precision}.$$

7 Example Search Strategy under H_{FD}

Under the fixed-dynamics hypothesis,

$$B^{(r)} = W^{(r)} \Theta^\top.$$

The unknowns are the shared interaction law Θ and the stage-dependent structural amplitudes $W^{(r)}$.

For fixed Θ , reconstruction in $W_{1:K}$ is linear. Define the profiled residual

$$E(\Theta) = \min_{W_{1:K}} \rho(B_{1:K}(W_{1:K}, \Theta)).$$

The nonlinear global search can then be carried out primarily in the lower-dimensional interaction-law space.

7.1 Feasible-Basin Search

Starting from an initial interaction law Θ_0 , use profiled multiscale block basin hopping. At iteration t :

1. Select a coefficient block

$$J \subseteq \{1, \dots, L\}.$$

2. Draw a multiscale perturbation

$$\Theta'_{\text{raw}} = \Theta_t + \delta, \quad \delta_J = \sigma D_J z,$$

where z is a random direction, σ is the jump scale, and D_J provides sensitivity scaling.

3. Starting from Θ'_{raw} , locally minimize the profiled objective $E(\Theta)$.
4. During each objective evaluation, solve

$$W_{1:K}^*(\Theta) = \arg \min_{W_{1:K}} \rho(B_{1:K}(W_{1:K}, \Theta)).$$

5. Let the resulting local minimum be Θ' . Accept an improving basin according to

$$E(\Theta') < E(\Theta_t).$$

6. Repeat until

$$E(\Theta) \leq \varepsilon.$$

This produces an observationally feasible representation

$$(W_{1:K}^*, \Theta) \in \mathcal{H}_{\text{FD}}^\varepsilon.$$

Random block size and multiscale jump amplitude allow single-coordinate, small-block, and global moves within one proposal family.

7.2 MDL Compression inside the Feasible Family

Once a feasible physical basin has been located, search proceeds over representations that remain inside

$$\mathcal{H}_{\text{FD}}^\varepsilon.$$

Candidate structural moves include

$$\boxed{\text{edge add,} \quad \text{edge drop,} \quad \text{edge swap,}}$$

while interaction-law moves include

$$\boxed{\text{atom add,} \quad \text{atom drop,} \quad \text{atom swap.}}$$

For each candidate move:

1. Modify the physical representation.
2. Refit the associated continuous coefficients.
3. Test whether

$$B_{1:K}(W, \Theta) \in \mathcal{B}_\varepsilon.$$

4. Evaluate

$$\Delta L_{\text{struct}}, \quad \Delta L_{\text{expr}}, \quad \Delta L_{\text{prec}}.$$

5. Combine them as

$$\boxed{\Delta L_{\text{MDL}} = \Delta L_{\text{struct}} + \Delta L_{\text{expr}} + \Delta L_{\text{prec}}.}$$

A candidate with

$$\Delta L_{\text{MDL}} < 0$$

gives a shorter admissible physical description.

7.3 Precision Profiling

For a fixed structural and expression model, first determine the continuous optimum. Let its coefficient vector be c^* , its minimum squared residual be R^* , and let

$$T = \varepsilon^2 \|y\|_2^2$$

be the allowed squared-error threshold.

If the relevant local design matrix has Gram matrix

$$G = A^\top A,$$

then the remaining quantization budget is

$$T - R^*.$$

A quantized coefficient vector $\tilde{c}$ remains feasible whenever

$$(\tilde{c} - c^*)^\top G(\tilde{c} - c^*) \leq T - R^*.$$

The smallest quantization depth satisfying this condition determines q^* and therefore L_{prec} .

Search proposals may prioritize low-order atoms, large residual correlations, large conditional structural gains, weak active edges, or multiscale block moves. These proposal mechanisms guide exploration, while the final selection criterion remains

$$L_{\text{MDL}}$$

subject to observational feasibility.

8 Complete TIDES Algorithm

The complete three-layer architecture is

Step 1: Temporal segmentation	$x(t) \longrightarrow \tau_{1:K-1},$
Step 2: Edge-space reconstruction	$(x(t), \tau_{1:K-1}, \Psi) \longrightarrow \mathcal{B}_\varepsilon,$ $\mathcal{B}_\varepsilon \longrightarrow \Delta\mathcal{B}_\varepsilon,$
Step 3: Physical compression	$(\mathcal{B}_\varepsilon, \Delta\mathcal{B}_\varepsilon, H) \longrightarrow \mathcal{H}^\varepsilon$ $\longrightarrow (W, \Theta)_H^*.$

The central lifting relation is

$$B_{m\ell}^{(r)} = w_m^{(r)} \theta_\ell^{(r)},$$

which converts topology–dynamics products into linearly reconstructable edge-space coefficients.

The resulting architecture separates three objects:

$$\begin{aligned} &\text{temporal structure} \longrightarrow \text{edge-space observational family} \\ &\quad \longrightarrow \text{physical minimum-description representation} \end{aligned}$$